

EDUCATION

Ph.D. Student, Statistics and Data Science

The Wharton School, University of Pennsylvania

2023 - 2027

GPA: 3.91/4.00

Bachelor of Science

University of Michigan, Ann Arbor

2019 - 2023

GPA: 3.89/4.00

Majors: Quadruple major in Statistics, Honors Mathematics, Economics, Data Science

Highest Honors in Statistics, Honors in Mathematics

PUBLICATIONS

*: Equal contribution

1. Learning Mixtures of Markov Chains and MDPs.

Chinmaya Kausik, Kevin Tan, Ambuj Tewari

ICML 2023, Short Oral Presentation

2. Offline Policy Evaluation and Optimization under Confounding

Kevin Tan, Yangyi Lu*, Chinmaya Kausik*, Yixin Wang, Maggie Makar, Ambuj Tewari*

AISTATS 2024

3. Informing Policy via Dynamic Models: Cholera in Haiti

Jesse Wheeler, Anna Rosengart, Zhuoxun Jiang, Kevin Tan, Noah Treutle, Edward Ionides

PLOS Comp Bio

4. A Natural Extension To Online Algorithms For Hybrid RL With Limited Coverage

Kevin Tan, Ziping Xu**

Reinforcement Learning Conference 2024

5. Hybrid RL Breaks Sample Size Barriers in Linear MDPs

Kevin Tan, Wei Fan, Yuting Wei

NeurIPS 2024

6. Distributed Least Squares in Small Space via Sketching and Bias Reduction

Sachin Garg, Kevin Tan, Michał Dereziński

NeurIPS 2024

7. Leveraging Offline Data in Linear Latent Bandits

Chinmaya Kausik, Kevin Tan, Ambuj Tewari

ICML 2025

8. Actor Critics Can Achieve Optimal Sample Efficiency

Kevin Tan, Wei Fan*, Yuting Wei*

ICML 2025

IN SUBMISSION

1. Automatic Differentiation Accelerates Inference for Partially-observed Markov Processes

R&R, JRSS-B

Kevin Tan, Giles Hooker, Edward L. Ionides

2. Learning Diffusion Policies with Sublinear Regret

In Submission, NeurIPS 2025

Kevin Tan, Zhihan Huang*, Yuting Wei*

3. Statistical Inference for Gradient Boosting Regression

In Submission, NeurIPS 2025

Haimo Fang, Giles Hooker, Kevin Tan

WORK EXPERIENCE

Applied Scientist Intern

Summer 2025 Till Graduation

Amazon

- Developing physics-informed behavior cloning models to analyze worker decision-making and improve warehouse throughput.
- Created prioritization scores for alarms within fulfillment centers via estimating heterogeneous treatment effects.

Machine Learning Research Engineer

Winter 2023, Summer 2023

Shade Inc.

- Developed various AI/ML products, including a text-audio search engine powered by multimodal embeddings, a similar search engine for 3D textures, jersey number recognition, fast online face clustering for facial recognition, lightweight asset description generation from a small set of tags, and an assortment of deep classifiers.
- Devised a framework for fast and lightweight fine-tuning of embeddings for large multimodal models.
- Oversaw a team that distilled BLIP, speeding up inference by over 7 times while maintaining performance.

Multi-asset, Mass Customized Solutions Intern

Summer 2020, 2021

Schroders

- Designed and developed an engine prototype for a digital multi-asset retirement and investment solution.
- Devised a recommendation algorithm for personalized user portfolios, via solving a sparse optimization problem with knapsack search heuristics, achieving portfolios 90% sparser than before and reducing unnecessary trades by 87%.

Lieutenant

2017-2018

Singapore Armed Forces, 18th Divisional Air Defence Battalion

- Air warfare officer in ground-based air defence, trained on a V-SHORAD system and short-ranged radar.
- Spearheaded projects and mentored personnel as a staff officer in the battalion's intelligence cell.

OPEN SOURCE SOFTWARE

pypomp

- Wrote the core engine behind a Python package for filtering through and training highly nonlinear and non-Gaussian state-space models. The package features a new training algorithm and gradient estimates for the particle filter.
- Achieved a 16x speedup over the **pomp** R package through just-in-time compilation and GPU acceleration.

Programming Python, R, C++, SQL, Git, JavaScript, Matlab.

Frameworks PyTorch, Tensorflow, Keras, JAX, PyMC3, Pandas, NumPy, SciPy, Scikit-Learn, Matplotlib, Tidyverse, Gurobi, CVXPY, Slurm, Aladdin, Google Data Studio.

Interests Building earphones, building guitars, cooking, baking.